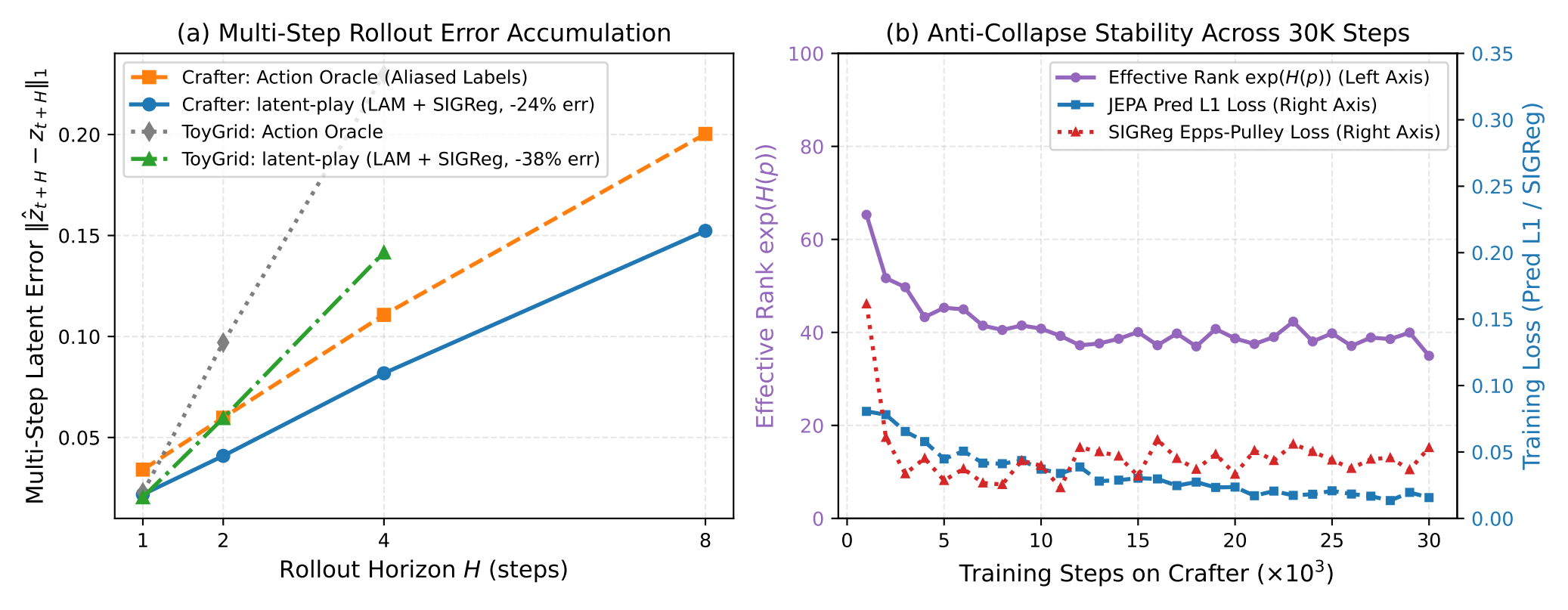
(a) Multi-Step Rollout Error Accumulation
Crafter: Action Oracle (Aliased Labels)
Crafter: latent-play (LAM + SIGReg, -24% err)
ToyGrid: Action Oracle
ToyGrid: latent-play (LAM + SIGReg, -38% err)
Multi-Step Latent Error $\|\hat{z}_{t+H} - z_{t+H}\|_1$
0.05
0.10
0.15
0.20
1
2
4
8
Rollout Horizon H (steps)
(b) Anti-Collapse Stability Across 30K Steps
Effective Rank exp($H(p)$) (Left Axis)
JEPA Pred L1 Loss (Right Axis)
SIGReg Epps-Pulley Loss (Right Axis)
Effective Rank exp($H(p)$)
0
20
40
60
80
100
Training Loss (Pred L1 / SIGReg)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0
5
10
15
20
25
30
Training Steps on Crafter (×10^3)